

essential to manage the complexities of multi-cloud and hybrid cloud environments.

- **Integration:** Integrating different cloud platforms and on-premises systems can be challenging and requires careful planning.
- **Cost management:** Accurately tracking and managing costs across multiple cloud environments can be complex.
- **Performance:** Ensuring optimal performance and latency in multi-cloud and hybrid cloud environments can require careful optimisation.

The role of open source in multi-cloud and hybrid cloud architectures

Open source software containerisation platforms have become an integral

component of modern IT environments, particularly in the context of multi-cloud and hybrid cloud architectures. These flexible and scalable platforms offer businesses greater agility, cost-effectiveness, and resilience and play a crucial role in the successful implementation and management of multi-cloud and hybrid cloud environments.

By providing a foundation of cost-effective, flexible, and community-supported tools, open source containerisation software empowers businesses to leverage the benefits of these innovative architectures, offering a comprehensive suite of solutions for building and operating multi-cloud and hybrid cloud environments.

As can be seen in Table 2, open source containerisation software offers significant benefits for businesses

adopting multi-cloud and hybrid cloud architectures. By reducing costs, enhancing flexibility, and fostering innovation, open source technologies enable businesses to optimise their IT operations and achieve strategic goals. Open source containerisation software is highly customisable and adaptable, making it suitable for complex environments. The community-driven nature of open source development provides valuable resources and support. Additionally, open source software can help businesses reduce vendor lock-in and negotiate better terms with vendors.

Steps to follow when implementing containerisation in a multi-cloud environment

Choose a container platform: Select a container platform like Docker or

	Virtualisation	Containerisation
Pros	- Improved resource utilisation over traditional deployment model - Application is isolated contrary to a traditional deployment model	- Quick boot up as is lightweight - Highly portable across host platforms - Relatively inexpensive - Highly scalable
Cons	- Slower to boot up and large OS image footprint - Limited scalability - Applications are not portable - Relatively more expensive	- Proper design and management are critical as containers can be spun faster - Applications are not fully isolated; security is a concern

Table 3: Pros and cons of containerisation and virtualisation

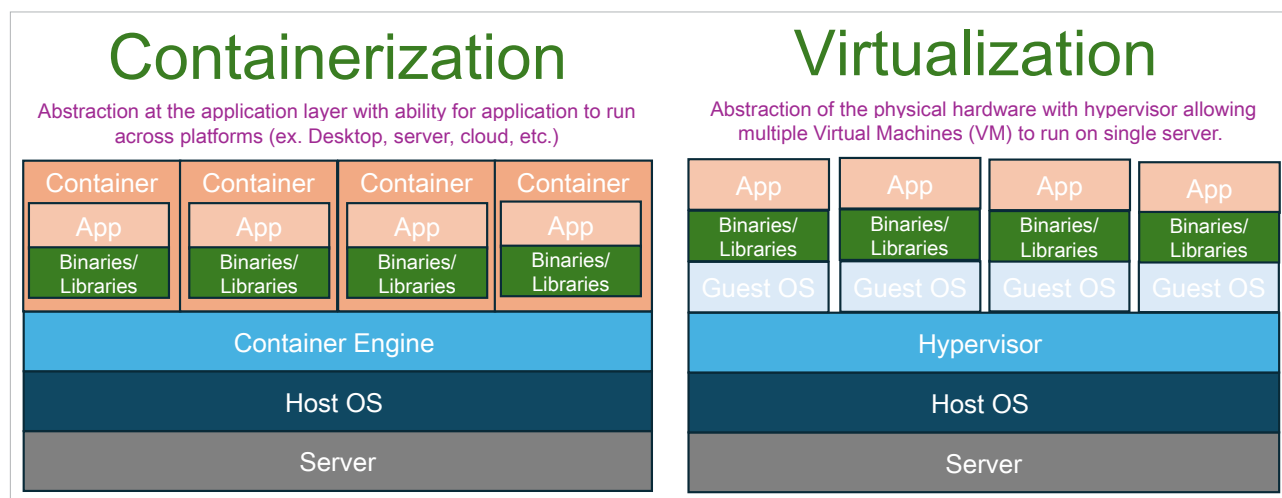


Figure 1: Containerisation vs virtualisation

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